

## DSA06 – Assessment Tool III

### Interactive Dashboard Project – Complete Tableau Practical Answers

Below is an **exam-ready solution for all 5 questions**, including Tableau construction steps, calculated fields, dashboard actions, interpretations, and design critique.

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#### Q1. Building a Sales Performance Dashboard

##### Objective

Create an interactive dashboard containing:

1. Sales by Region – Bar Chart
2. Sales trend – Line Chart
3. Sales by Region – Map
4. Global Region Filter

##### Given Data

| Region Category |                 | Sales  |
|-----------------|-----------------|--------|
| North           | Furniture       | 32,000 |
| South           | Furniture       | 28,500 |
| East            | Furniture       | 41,000 |
| West            | Furniture       | 25,500 |
| North           | Technology      | 51,000 |
| South           | Technology      | 47,500 |
| East            | Technology      | 62,000 |
| West            | Technology      | 39,500 |
| North           | Office Supplies | 21,000 |
| South           | Office Supplies | 19,500 |
| East            | Office Supplies | 27,000 |
| West            | Office Supplies | 17,500 |

##### Tableau Steps

## 1. Connect Data

- Open Tableau.
- Select **Microsoft Excel/Text File** depending on how the data is stored.
- Enter the data with fields:
  - Region
  - Category
  - Sales

Set **Sales** as a numerical measure.

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## 2. Bar Chart – Sales by Region

Create worksheet: **Sales by Region**

- Drag Region → **Columns**
- Drag Sales → **Rows**
- Select **Bar Chart**
- Drag Category → **Color**
- Sort regions by descending Sales.
- Enable data labels.

### Expected regional totals

#### Region Total Sales

East 130,000

North 104,000

South 95,500

West 82,500

Therefore:

**East > North > South > West**

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## 3. Map – Sales by Region

Create worksheet: **Regional Sales Map**

- Double-click Region.
- Tableau should recognize the region names only if geographic roles are appropriate.
- If Tableau does not recognize these generic North, South, East, and West values as geographic locations, a true geographic map cannot be created from this dataset alone.

For an academic practical, the intended design can be represented as a **symbol/map-style regional view**, but do **not fabricate geographical coordinates**.

Use:

- Sales → Size
  - Sales → Color
  - Region → Detail
- 

#### 4. Line Chart

There is an important issue with the supplied dataset:

**The dataset contains no Month/Date field.**

Therefore, a genuine **Sales trend by Month cannot be constructed without inventing data**, which would violate the instruction to use the given data exactly.

**Correct approach in the practical**

State:

“The provided dataset does not contain a Month or Date field. Hence, a genuine monthly sales trend cannot be generated without additional time information. I have retained the original data and used the available dimensions for the remaining visualizations.”

If the faculty expects a line chart despite this issue, use **Category on Columns and Sales on Rows**, but label it clearly as **Sales by Category**, not “Monthly Sales Trend.”

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#### 5. Global Region Filter

On the bar chart:

- Drag Region → **Filters**

- Select **All**
- Right-click the filter → **Show Filter**
- Add the worksheets to the dashboard.
- Select the Region filter menu.
- Choose:

### Apply to Worksheets → All Using This Data Source

Now selecting **East**, for example, updates all connected views.

## 6. Dashboard Layout

Suggested layout:

|       |                                          |  |                    |
|-------|------------------------------------------|--|--------------------|
| ----- |                                          |  |                    |
|       | SALES PERFORMANCE DASHBOARD              |  |                    |
| ----- |                                          |  |                    |
|       | Region Filter                            |  |                    |
| ----- |                                          |  |                    |
|       | Sales by Region                          |  | Regional Sales Map |
|       |                                          |  |                    |
|       | BAR CHART                                |  | MAP                |
| ----- |                                          |  |                    |
|       | Sales by Category / Available Trend View |  |                    |
|       | LINE CHART                               |  |                    |
| ----- |                                          |  |                    |

## Analysis

The **East region** has the highest total sales at **₹130,000**, while West has the lowest at **₹82,500**.

The strongest region-category combination is:

**East – Technology = ₹62,000**

The next strongest is:

North – Technology = ₹51,000

Technology is the strongest category in every region. The East region consistently performs best across all three categories.

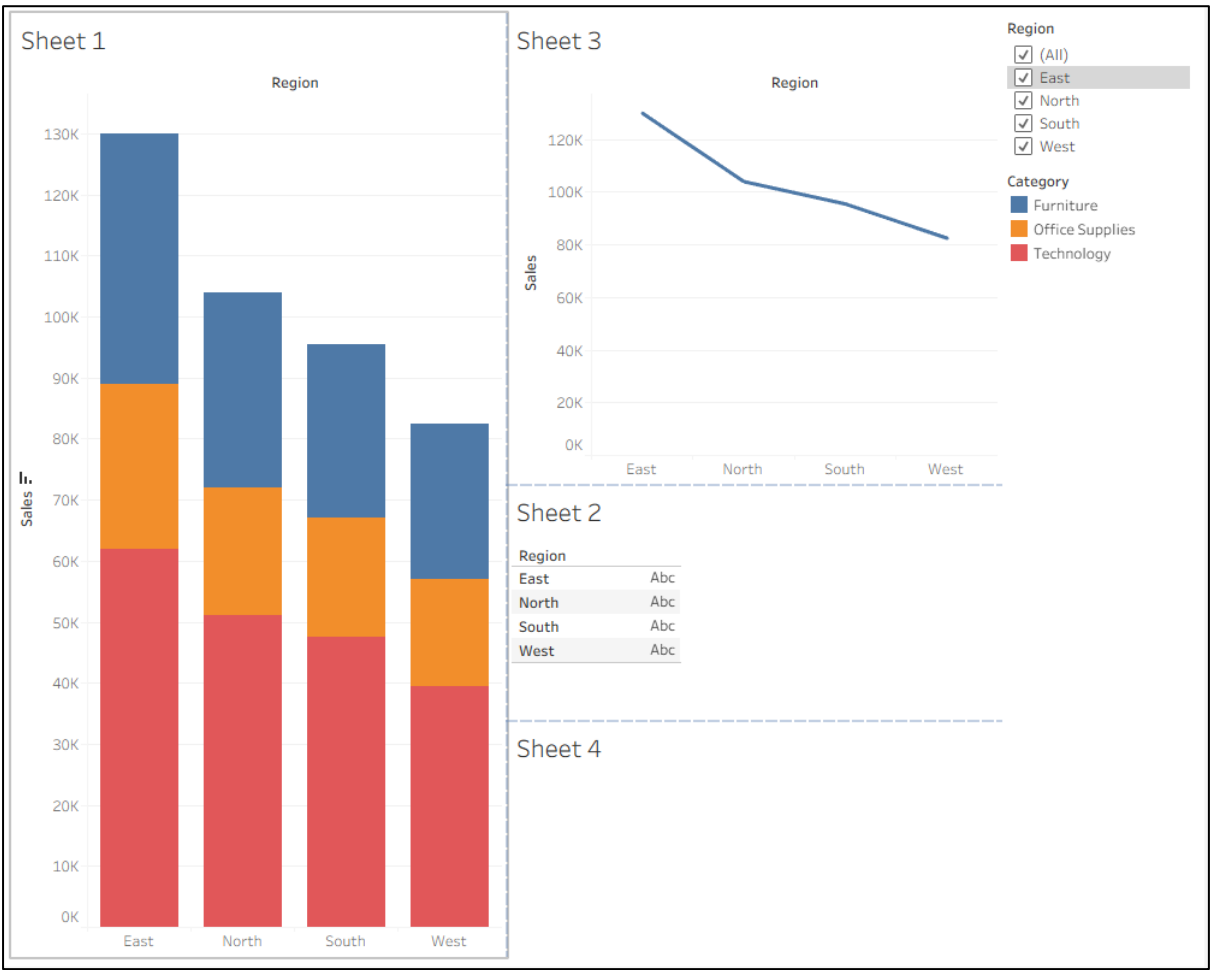
Category totals

| Category        | Total Sales |
|-----------------|-------------|
| Furniture       | 127,000     |
| Technology      | 200,000     |
| Office Supplies | 85,000      |

Therefore, Technology is the major sales driver.

Dashboard Design Interpretation

The dashboard supports quick comparison because the bar chart gives an immediate ranking of regions, while the map provides regional visual context. The global Region filter allows the user to focus on a particular region without manually changing each visualization.



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## Q2. KPI Dashboard with Parameters

### Objective

Create:

- Average Rating KPI
  - Total Employees KPI
  - Average Attendance KPI
  - Department-wise bar chart
  - Parameter to switch between Rating and Attendance
- 

### 1. Calculate KPI Values

There are **10 employees**.

#### Average Rating

Ratings:

7.8, 8.1, 7.5, 7.9, 8.6, 8.9, 7.2, 7.4, 8.0, 8.3

Total:

**79.7**

Average:

**$79.7 / 10 = 7.97$**

#### Average Attendance

Attendance:

92, 95, 88, 90, 96, 97, 85, 87, 93, 94

Total:

**917**

Average:

**$917 / 10 = 91.7\%$**

### KPI Summary

| KPI                | Value        |
|--------------------|--------------|
| Average Rating     | <b>7.97</b>  |
| Total Employees    | <b>10</b>    |
| Average Attendance | <b>91.7%</b> |

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## Tableau Calculated Fields

### Average Rating KPI

Create calculated field:

AVG([Rating])

Format to 2 decimal places.

### Total Employees

COUNT([S.No])

### Average Attendance

AVG([Attendance])

Format as percentage if appropriate for the chosen representation.

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## 2. Create Parameter

Go to:

**Data → Create Parameter**

Name:

### Select Measure

Data Type:

### String

Allowable values:

Average Rating

Average Attendance

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### 3. Create Parameter-Driven Calculated Field

Create:

#### Selected Measure

IF [Select Measure] = "Average Rating" THEN

AVG([Rating])

ELSE

AVG([Attendance])

END

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### 4. Bar Chart

Create worksheet:

#### Department Performance

- Department → Columns
- Selected Measure → Rows
- Department → Color if desired
- Show data labels.

Display the parameter:

#### Select Measure → Show Parameter

Now the user can select:

#### Average Rating

or

#### Average Attendance

and the same chart changes dynamically.

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### Department Results

#### Average Rating

Department Average Rating

|             |      |
|-------------|------|
| Engineering | 8.75 |
| HR          | 8.15 |
| Sales       | 7.95 |
| Marketing   | 7.70 |
| Support     | 7.30 |

Average Attendance

Department Average Attendance

|             |      |
|-------------|------|
| Engineering | 96.5 |
| HR          | 93.5 |
| Sales       | 93.5 |
| Marketing   | 89.0 |
| Support     | 86.0 |

Dashboard Layout

|       |                                    |           |                |
|-------|------------------------------------|-----------|----------------|
| ----- |                                    |           |                |
|       | HR PERFORMANCE DASHBOARD           |           |                |
| ----- |                                    |           |                |
|       | Avg Rating                         | Employees | Avg Attendance |
|       | 7.97                               | 10        | 91.7%          |
| ----- |                                    |           |                |
|       | Select Measure: [Average Rating ▼] |           |                |
| ----- |                                    |           |                |
|       |                                    |           |                |
|       | Department Performance             |           |                |
|       |                                    |           |                |
|       | BAR CHART                          |           |                |

## Interpretation

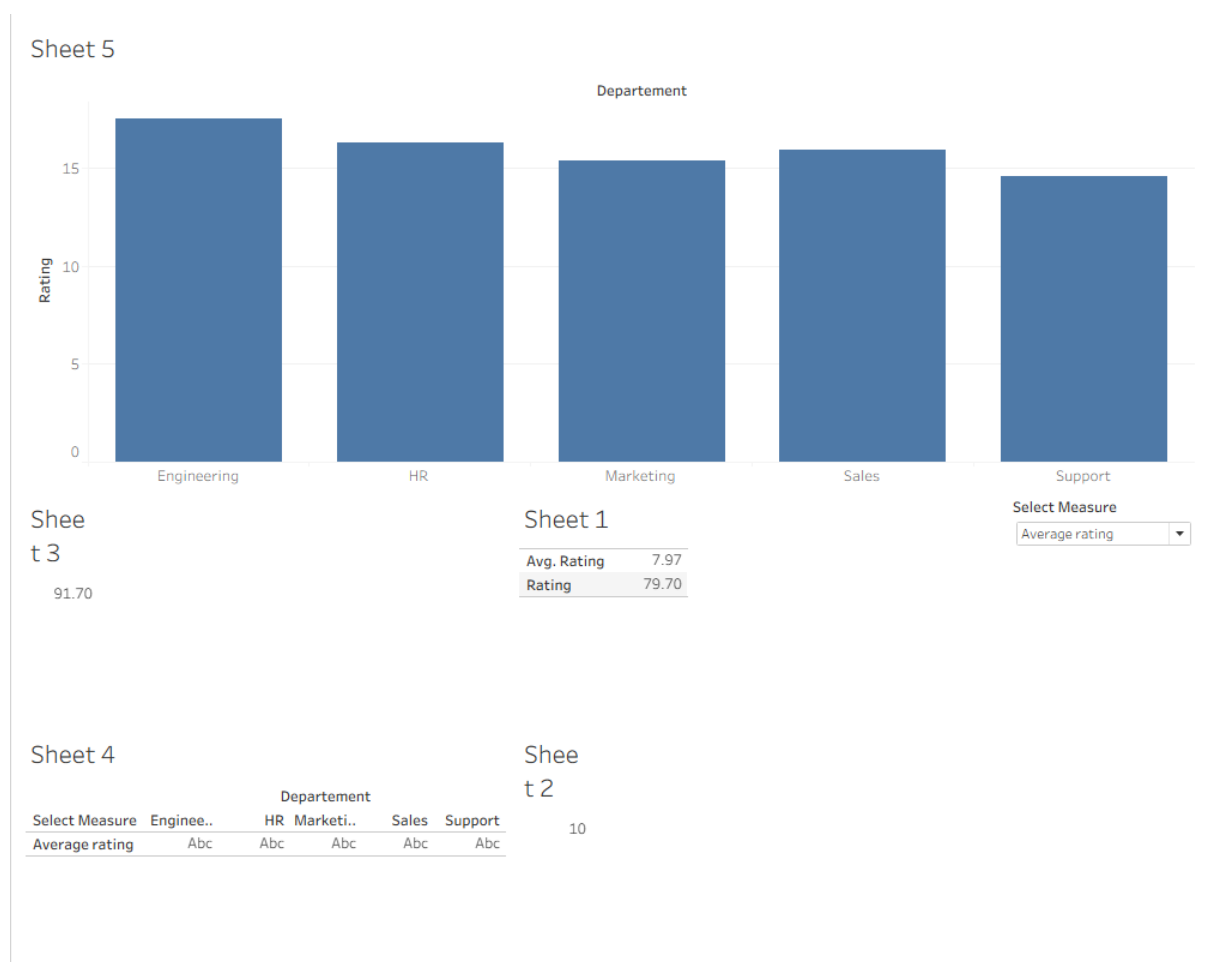
Engineering has the highest average rating (**8.75**) and highest average attendance (**96.5%**).

Support has the lowest rating (**7.30**) and attendance (**86%**).

## Why parameters are useful

Parameters make the dashboard flexible because one visualization can display different measures based on the user's selection. This reduces the need for multiple static charts and saves dashboard space.

A parameter therefore provides **interactive measure switching** and improves exploratory analysis.



## Q3. Drill-Down Dashboard with Dashboard Actions

## Objective

Create:

1. Program-level enrollment chart
  2. Specialization-level detail chart
  3. Filter action connecting the two
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### 1. Program-Level Summary

Create worksheet:

#### Program Enrollment

- Program → Columns
- Enrollment → Rows
- Select **Bar Chart**
- Add data labels.

#### Program totals

##### CSE

$$65 + 40 + 90 = \mathbf{195}$$

##### ECE

$$35 + 50 = \mathbf{85}$$

##### MECH

$$30 + 45 + 28 = \mathbf{103}$$

#### Program Enrollment

CSE      **195**

MECH    **103**

ECE      **85**

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### 2. Specialization Chart

Create worksheet:

## Specialization Enrollment

- Specialization → Columns
- Enrollment → Rows
- Select Bar Chart
- Add Program → Detail or Color.

Results:

| Program | Specialization | Enrollment |
|---------|----------------|------------|
| CSE     | AI/ML          | 65         |
| CSE     | Cyber Security | 40         |
| CSE     | Core CSE       | 90         |
| ECE     | VLSI           | 35         |
| ECE     | Communications | 50         |
| MECH    | Robotics       | 30         |
| MECH    | Thermal        | 45         |
| MECH    | Design         | 28         |

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## 3. Create Dashboard

Place both worksheets on one dashboard:

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|       |                              |  |
|-------|------------------------------|--|
|       | STUDENT ENROLLMENT DASHBOARD |  |
| ----- |                              |  |
|       |                              |  |
|       | PROGRAM ENROLLMENT           |  |
|       | CSE                          |  |
|       | MECH                         |  |
|       | ECE                          |  |
|       |                              |  |

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|  |                        |  |
|--|------------------------|--|
|  | SPECIALIZATION DETAILS |  |
|  |                        |  |
|  | AI/ML                  |  |
|  | Cyber                  |  |
|  | Core CSE               |  |

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#### 4. Configure Filter Action

Go to:

**Dashboard → Actions → Add Action → Filter**

Set:

- **Source Sheet:** Program Enrollment
- **Target Sheet:** Specialization Enrollment
- Run action on: **Select**
- Clearing selection: **Show all values**

Now clicking **CSE** filters the specialization chart to:

- AI/ML – 65
- Cyber Security – 40
- Core CSE – 90

Clicking **ECE** shows:

- VLSI – 35
- Communications – 50

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#### Interpretation

CSE has the highest total enrollment with **195 students**.

The largest individual specialization is:

**CSE – Core CSE = 90 students**

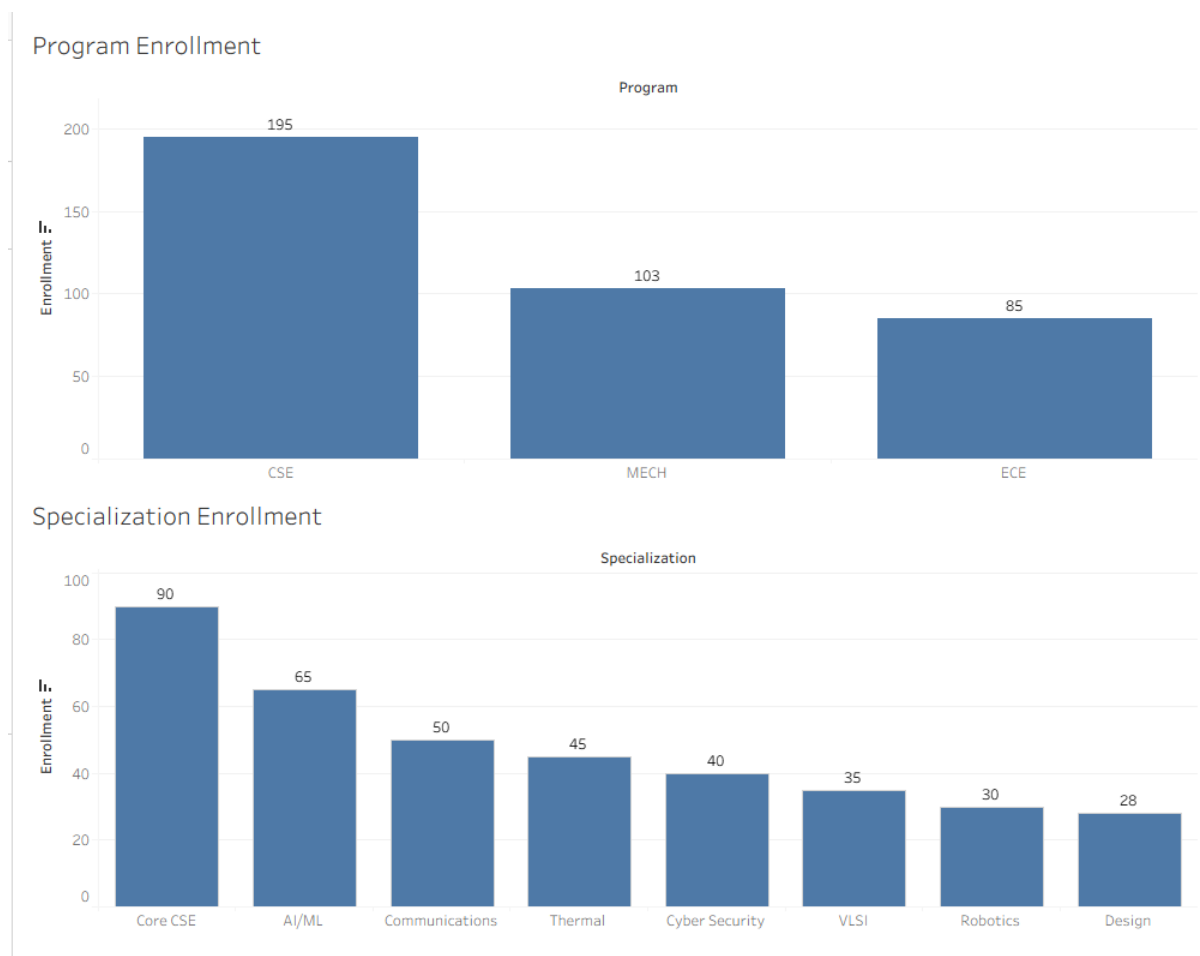
The drill-down interaction improves exploration because the dashboard initially presents a simple program-level summary. Users can then select a program to investigate only its relevant specializations.

### Advantages

- Reduces visual clutter
- Supports interactive exploration
- Makes the dashboard easier to understand
- Allows users to move from summary to detail

### Limitations

- Users may not discover the detail information unless they click a program.
- Comparing specializations across different programs becomes harder.
- The user must interact with the dashboard to see all details.
- Too many hierarchical levels can make navigation confusing.



## Q4. Multivariate Analysis Dashboard

### Objective

Create:

1. Advertising vs Sales scatterplot
  2. Website Visits represented using color/size
  3. Trend line
  4. Sales distribution chart
- 

### 1. Scatterplot

Create worksheet:

#### Advertising vs Sales

- Drag Advertising → Columns
- Drag Sales → Rows
- Change Marks type to **Circle**
- Drag Visits → **Size**
- Optionally drag Visits → **Color**

This creates a three-variable visualization.

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### 2. Add Trend Line

Go to:

**Analytics → Trend Line → Linear**

Drop it into the visualization.

The trend line should rise from left to right.

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### 3. Sales Distribution

Create second worksheet:

#### Sales Distribution

For example:

- Drag Sales → Columns
- Use **Histogram**

Alternatively, use a **Box Plot**.

Sales range from:

**48 to 86**

The distribution is fairly regular and has no obvious extreme outliers.

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#### 4. Dashboard Layout



## Relationship Analysis

The relationship between **Advertising expenditure and Sales is strongly positive**.

As advertising expenditure increases, sales also increase.

The values show a very consistent upward pattern:

- Advertising = 10 → Sales = 48
- Advertising = 20 → Sales = 61
- Advertising = 30 → Sales = 74
- Advertising = 38 → Sales = 86

The points lie very close to a straight increasing trend.

Therefore:

**Higher advertising expenditure is strongly associated with higher sales.**

However, correlation alone does not prove that advertising expenditure directly causes the increase in sales.

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## Website Visits Interpretation

Website Visits also increase as Advertising increases.

For example:

### Advertising Visits Sales

|    |     |    |
|----|-----|----|
| 10 | 120 | 48 |
| 20 | 195 | 61 |
| 30 | 275 | 74 |
| 38 | 340 | 86 |

Using color or size for Website Visits adds a third dimension and helps show that higher advertising levels are associated with more website visits.

### Does it create clutter?

For this small dataset, the encoding is useful because there are only **15 observations**.

However, using both **color and size simultaneously** could make the scatterplot unnecessarily complicated.

## Recommended design

Use:

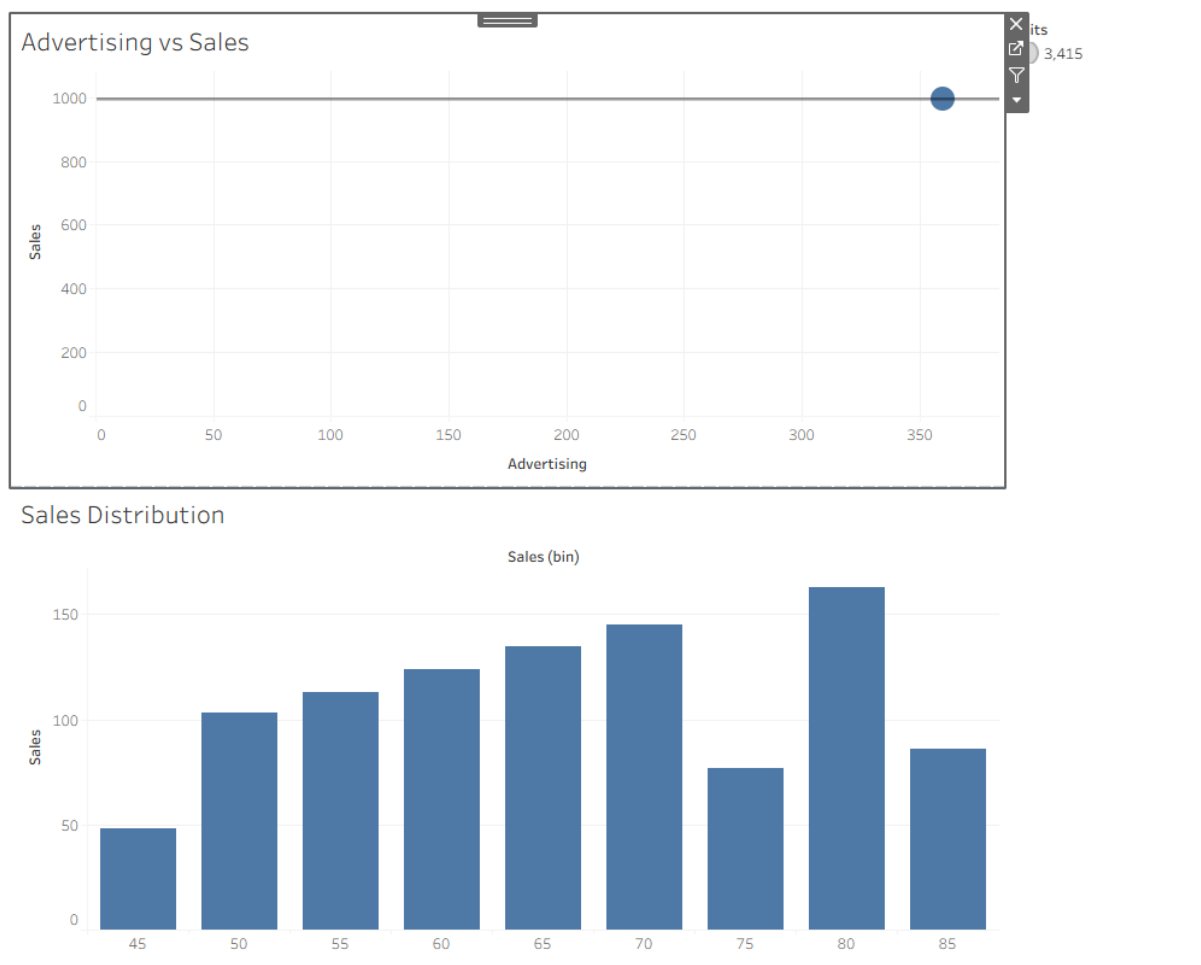
**Advertising = X-axis**

**Sales = Y-axis**

**Website Visits = Size**

and keep the color scheme simple.

This gives the viewer three variables without excessive visual clutter.



## Q5. Misleading Dashboard and Redesign

### Dataset

#### Department Pass Percentage

CSE 91

ECE 93

## Department Pass Percentage

|       |    |
|-------|----|
| EEE   | 92 |
| MECH  | 89 |
| CIVIL | 90 |
| AIDS  | 95 |
| IT    | 94 |

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### Part A – Misleading Dashboard

Create a bar chart with:

- Department → Columns
- Pass Percentage → Rows

Then intentionally introduce the requested misleading choices.

#### 1. Truncated Y-axis

Set the Y-axis approximately from:

**88 to 96**

instead of starting at zero.

This makes small differences appear much larger.

For example, the difference between:

- MECH = 89%
- AIDS = 95%

is actually only **6 percentage points**.

A truncated axis can visually make the difference appear dramatically larger.

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#### 2. Inappropriate Color Scheme

Use many unrelated bright colors for departments.

For example:

CSE → different color

ECE → different color

EEE → different color

MECH → different color

CIVIL → different color

AIDS → highly contrasting color

IT → different color

This makes the viewer focus on color differences rather than the actual values.

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### 3. Exaggerated Emphasis

Highlight **AIDS = 95%** with:

- Much stronger color
- Larger mark
- Different formatting

This can make AIDS appear overwhelmingly better than the other departments.

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## Part B – Redesigned Dashboard

Create a second dashboard using honest visualization principles.

### 1. Appropriate Baseline

Set the Y-axis baseline to:

**0%**

This represents the magnitude of the values accurately.

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### 2. Order Categories

Sort departments from highest to lowest:

**Rank Department Pass %**

|   |      |    |
|---|------|----|
| 1 | AIDS | 95 |
| 2 | IT   | 94 |

### Rank Department Pass %

|   |       |    |
|---|-------|----|
| 3 | ECE   | 93 |
| 4 | EEE   | 92 |
| 5 | CSE   | 91 |
| 6 | CIVIL | 90 |
| 7 | MECH  | 89 |

This makes ranking immediately visible.

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### 3. Consistent Color Encoding

Use one consistent color for all departments, or use a restrained sequential scale if there is a meaningful reason to encode magnitude.

Do not assign unrelated colors to every department.

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### 4. Clear Labels

Enable:

#### Show Mark Labels

Display:

AIDS – 95%

IT – 94%

ECE – 93%

EEE – 92%

CSE – 91%

CIVIL – 90%

MECH – 89%

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### Comparison of the Two Dashboards

| Feature           | Misleading Dashboard       | Redesigned Dashboard |
|-------------------|----------------------------|----------------------|
| Y-axis            | Truncated                  | Starts at 0          |
| Category order    | Unordered                  | Ordered              |
| Colors            | Inconsistent               | Consistent           |
| Highlighting      | Exaggerated                | Neutral              |
| Labels            | Poor/limited               | Clear                |
| Interpretation    | Can exaggerate differences | More accurate        |
| Viewer perception | Potentially misleading     | Honest and clear     |

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### Q5 Interpretation

The actual difference between the highest and lowest department is:

**95% – 89% = 6 percentage points**

Although AIDS has the highest pass percentage, the departments are actually relatively close to one another.

The misleading dashboard can make the differences look much larger because the y-axis begins near the observed values rather than at zero. An inappropriate color scheme and excessive emphasis on AIDS can further influence the viewer's perception.

The redesigned dashboard communicates the data more honestly because it uses a consistent baseline, ordered categories, restrained color encoding, and explicit labels.

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### Final Dashboard Design Principles Used

| Principle                   | Application                            |
|-----------------------------|----------------------------------------|
| Appropriate chart selection | Bar, line, map, scatterplot, histogram |
| Consistent encoding         | Same visual meaning throughout         |
| Clear labels                | Values and titles displayed            |
| Appropriate baseline        | Zero baseline where magnitude matters  |
| Interactivity               | Filters, parameters and actions        |

## Principle

## Application

Drill-down

Summary → detailed information

Avoid visual clutter

Limited colors and unnecessary elements

Honest representation

No exaggerated visual differences

Comparison

Ordered categories and aligned scales

User control

Global filters and parameters

